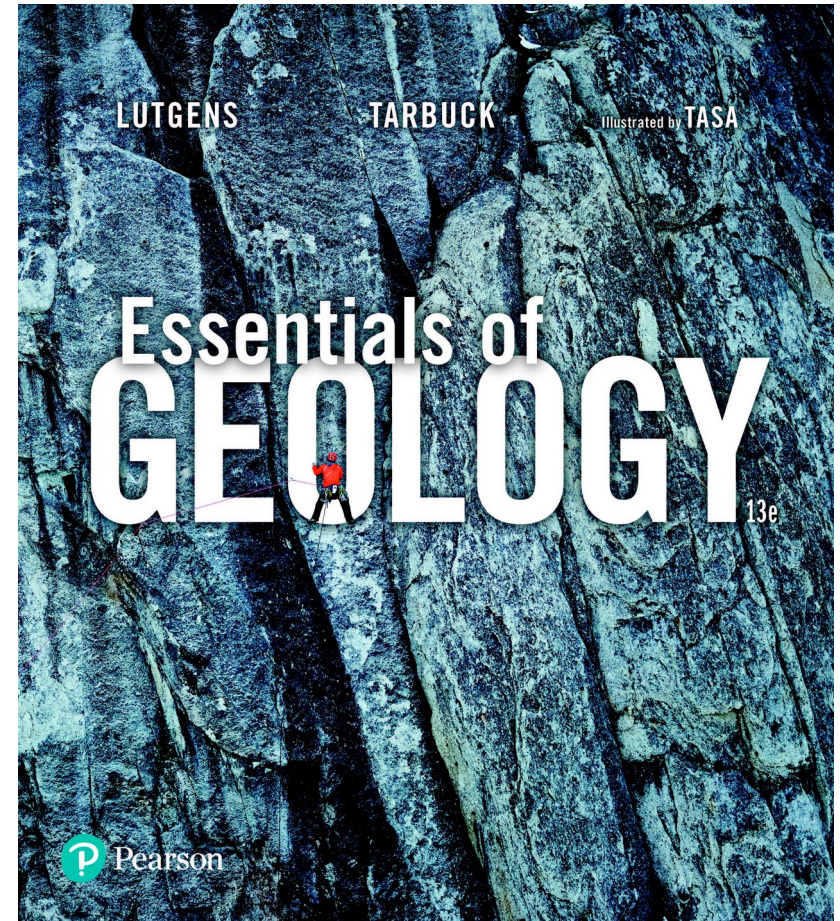


Lecture 6 Outline

Essentials of Geology

Thirteenth Edition



Weathering & Soils

Merry Wilson
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Learning Objectives

- **LO 6.1** Define *weathering* and distinguish between the two main categories of weathering.
- **LO 6.2** List and describe four examples of mechanical weathering.
- **LO 6.3** Discuss the importance of water and carbonic acid in chemical weathering processes.
- **LO 6.4** Summarize the factors that influence the type and rate of rock weathering.

Learning Objectives (cont.)

- **LO 6.5** Define *soil* and explain why soil is referred to as an *interface*. List and briefly discuss five controls of soil formation.
- **LO 6.6** Sketch, label, and describe an idealized soil profile. Explain the need for classifying soils.
- **LO 6.7** Explain the detrimental impact of human activities on soil.
- **LO 6.8** Relate weathering to the formation of certain ore deposits.

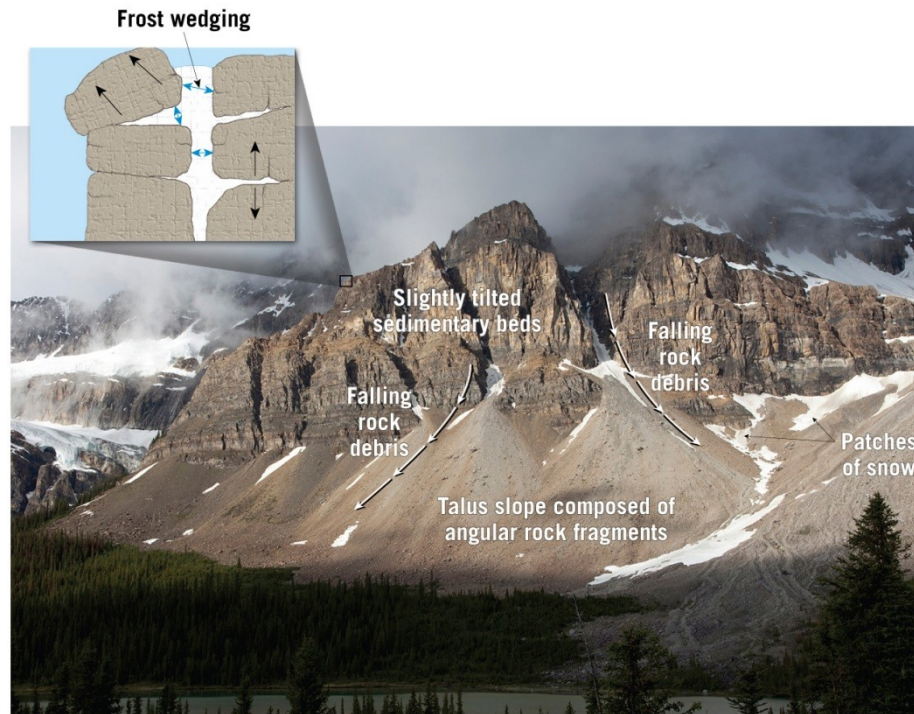
6.1 Weathering

- Weathering: The physical breakdown and chemical alteration of rock at or near Earth's surface
 - Mechanical weathering: Physical forces that break rock into smaller pieces without changing composition
 - Chemical weathering: Chemical transformation into new compounds

6.2 Mechanical Weathering

- Types of mechanical weathering
 - Frost wedging: Alternate freezing and thawing of water in fractures and cracks promotes the disintegration of rocks.

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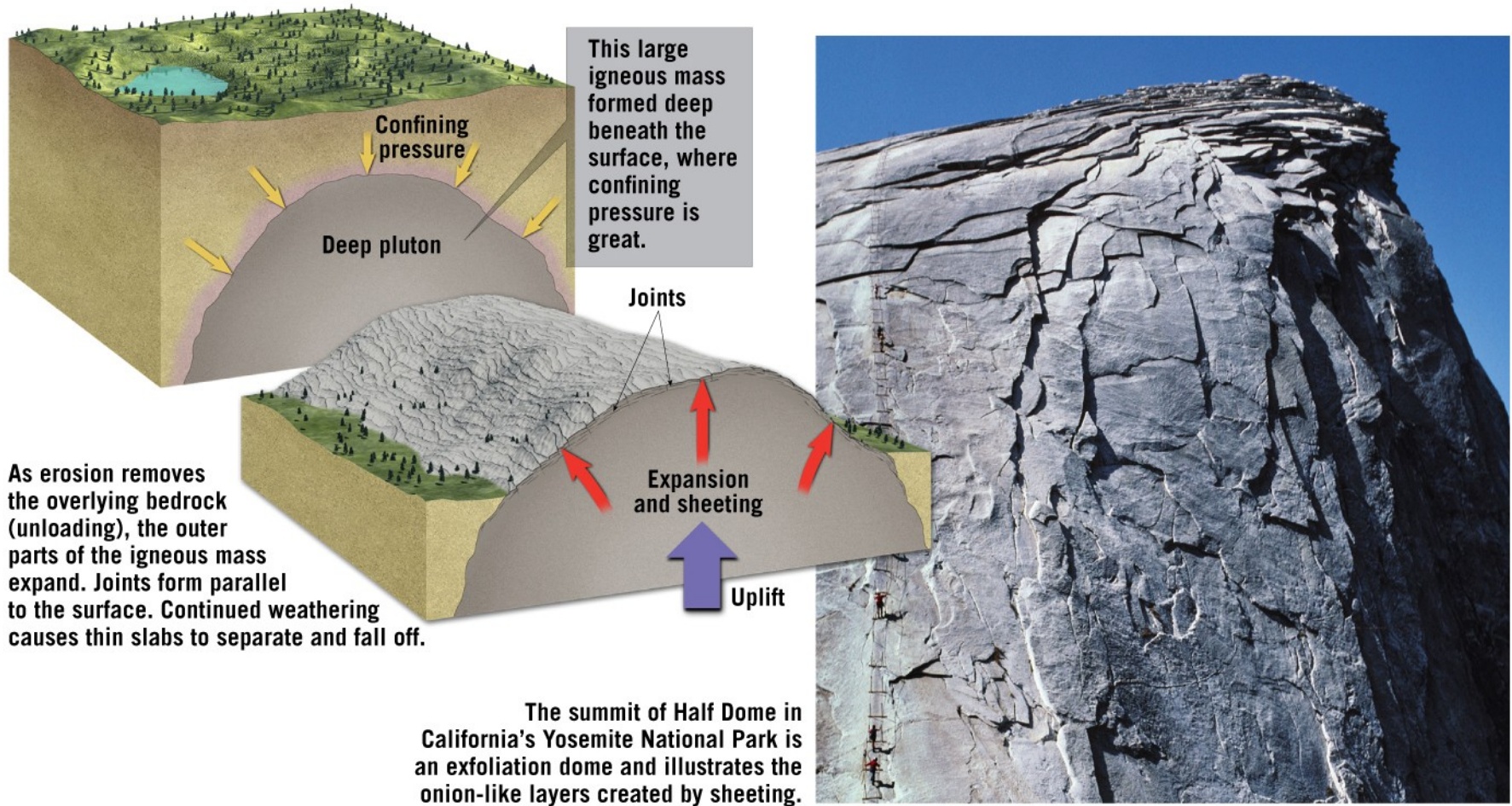


6.2 Mechanical Weathering (cont.)

- Mechanical weathering (*continued*)
 - Sheetting: Exfoliation of igneous and metamorphic rocks at the Earth's surface as a result of a reduction in confining pressure
 - Salt crystal growth
 - Sea spray or salty groundwater penetrates crevices and pore spaces in rocks
 - As the water evaporates, salt crystals form and enlarge the crevices
 - Biological activity: Disintegration resulting from plants and animals

Exfoliation of Igneous Rocks

Tutorial



6.3 Chemical Weathering

- Chemical weathering
 - Breaks down rock components and the internal structures of minerals
 - Most important agent involved in chemical weathering is water (responsible for transport of ions and molecules involved in chemical processes)

6.3 Chemical Weathering (cont.)

- Major processes of chemical weathering
 - Carbonic Acid
 - Carbon dioxide is dissolved in water and creates carbonic acid
 - Dissolves carbonate rocks
 - Oxidation
 - Any chemical reaction in which a compound or radical loses electrons
 - Important in decomposing ferromagnesian minerals

6.3 Chemical Weathering (cont.)

- Major processes of chemical weathering
 - How granite weathers
 - Chemically attacked by carbonic acid
 - Hydrogen ion attacks and replaces other positive ions, like potassium
 - Most abundant product is clay minerals

Products of Chemical Weathering

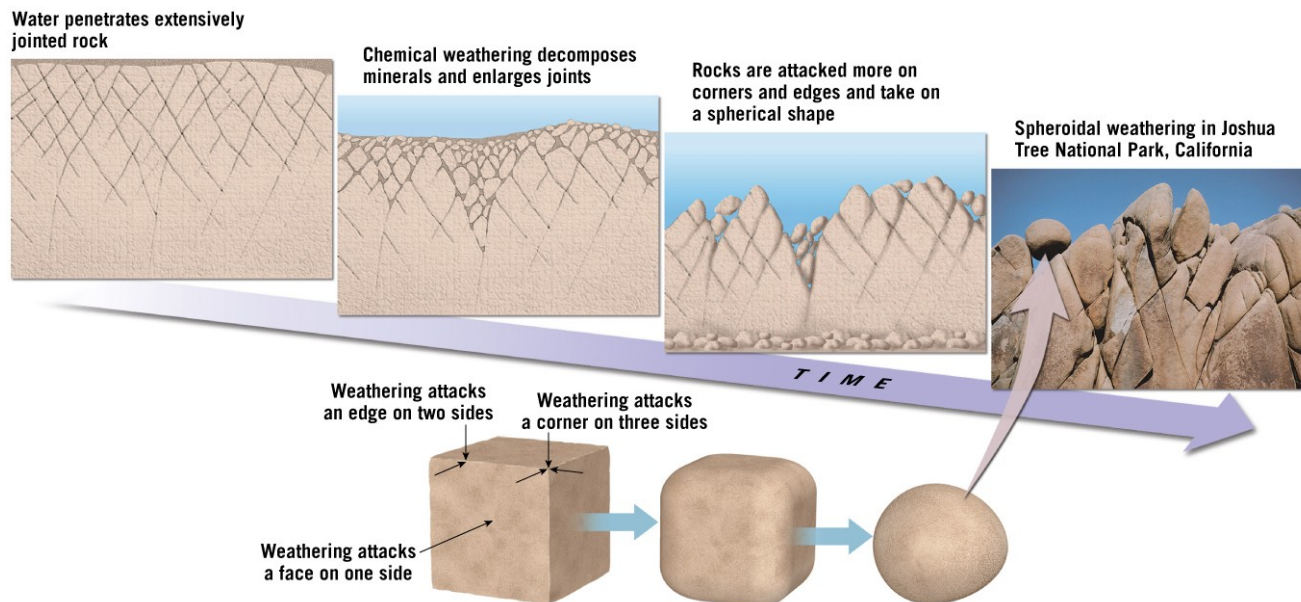
Table 6.1 Products of Chemical Weathering

Mineral	Residual Products	Material in Solution
Quartz	Quartz grains	Silica
Feldspars	Clay minerals	Silica, K^+ , Na^+ , Ca^{2+}
Amphibole (hornblende)	Clay minerals	Silica, Ca^{2+} , Mg^{2+}
	Limonite	
	Hematite	
Olivine	Limonite	Silica
	Hematite	Mg^{2+}

6.3 Chemical Weathering

- Alterations caused by chemical weathering
 - Decomposition of unstable minerals
 - Generation or retention of materials that are stable
 - Physical changes such as the rounding of corners or edges (spheroidal weathering)

[Tutorial](#)



6.4 Rates of Weathering

- Factors affecting weathering
 - Rock characteristics
 - Rocks containing calcite (marble and limestone) readily dissolve in weakly acidic solutions.
 - Silicate minerals weather in the same order as their order of crystallization.

[Tutorial](#)

This granite headstone was erected in 1868. The inscription still looks fresh.



This headstone of calcite-rich marble dates from 1874, six years after the granite stone. The inscription is barely legible.



6.4 Rates of Weathering (cont.)

- Others factors affecting weathering
 - Climate
 - Temperature and moisture are the most crucial factors.
 - Chemical weathering is most effective in areas of warm, moist climates.
 - Differential weathering
 - Masses of rock do not weather uniformly because of regional and local factors.
 - Results in many unusual and spectacular rock formations and landforms.

Differential Weathering

[Mobile Field Trip](#)



6.5 Soil: An Indispensable Resource

- Soil is a combination of mineral and organic matter, water, and air.
 - That portion of the regolith (rock and mineral fragments produced by weathering) that supports the growth of plants.

Typical Components in a Soil That Yield Good Plant Growth



6.5 Soil: An Indispensable Resource

- Factors controlling soil formation
 - Parent material
 - Residual soil: Parent material is the underlying bedrock
 - Transported soil: Forms in place on parent material that has been carried from elsewhere and deposited

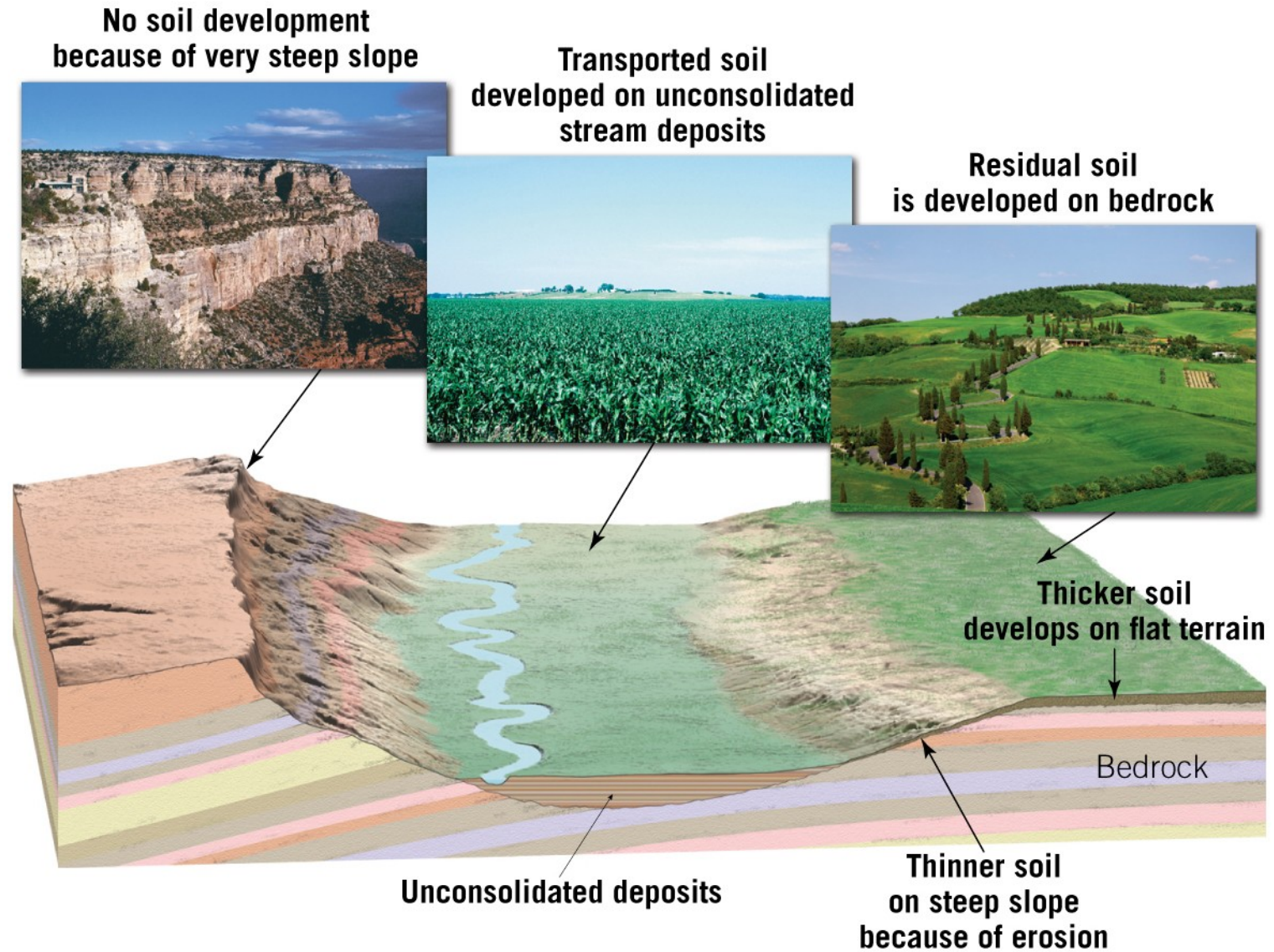
6.5 Soil: An Indispensable Resource (cont.)

- Factors controlling soil formation
 - Time
 - Important in all geologic processes
 - Amount of time for soil formation varies for different soils depending on geologic and climatic conditions
 - Climate
 - Most influential control of soil formation
 - Key factors are temperature and precipitation

6.5 Soil: An Indispensable Resource (cont.)

- Factors controlling soil formation
 - Plants and animals
 - Organisms influence the soil's physical and chemical properties
 - Also furnish organic matter to the soil
 - Topography
 - Steep slopes often have poorly developed soils
 - Optimum terrain is a flat-to-undulating upland surface

Soil Development



6.6 Describing & Classifying Soils

- The soil profile
 - Soil forming processes operate from the surface downward.
 - Vertical differences are called horizons, which are zones or layers of soil.

6.6 Describing & Classifying Soils (cont.)

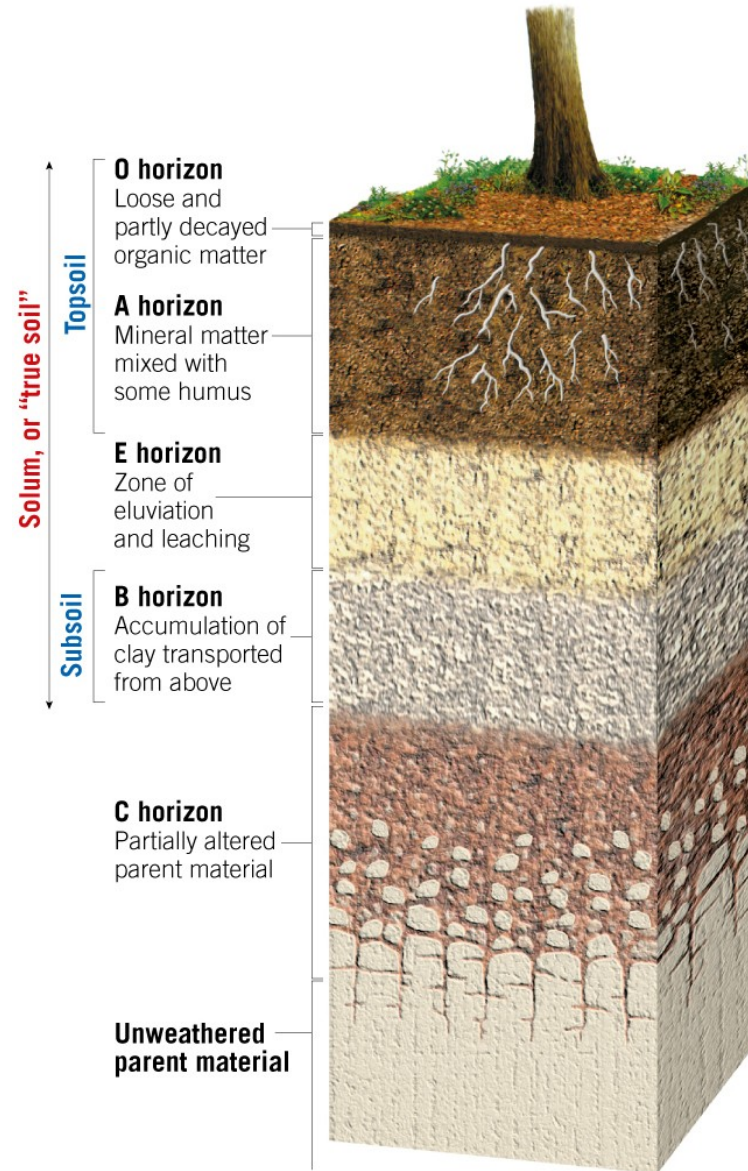
- The soil profile
 - O horizon: Organic matter
 - A horizon: Organic and mineral matter
 - High biological activity
 - Together the O and A horizons make up the topsoil
 - E horizon: Little organic matter
 - Zone of eluviation and leaching

6.6 Describing & Classifying Soils (cont.)

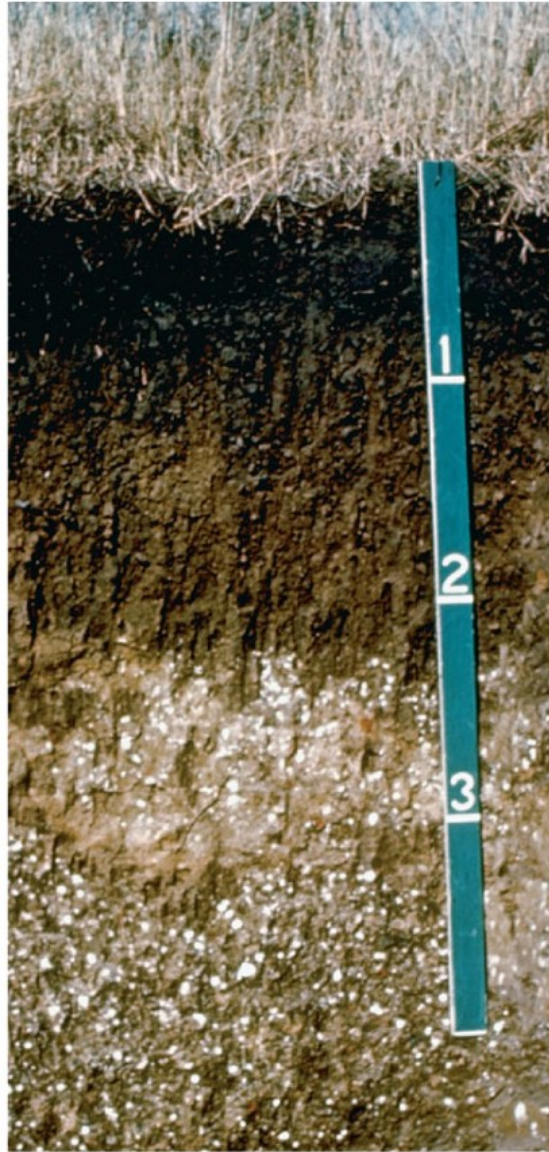
- The soil profile
 - *B* horizon: Zone of accumulation
 - *C* horizon: Partially altered parent material
- The *O*, *A*, *E*, and *B* horizons together are called the solum, or “true soil.”

An Idealized Soil Profile

Tutorial



A Soil Profile Showing Different Horizons



6.6 Describing & Classifying Soils

- Soil types
 - Groups consist of items that have certain important characteristics in common
- Soils are classified using a system known as the soil taxonomy.
 - Based on physical and chemical properties of the soil profile

6.6 Describing & Classifying Soils (cont.)

- Soil taxonomy
 - Includes six hierarchical categories of classification, ranging from order (broadest) to series (most specific)
 - Especially useful for agricultural and related land-use purposes

6.7 The Impact of Human Activities on Soil

- The agricultural productivity of soils can be improved through fertilization and irrigation.
- Soils can be damaged or destroyed by careless activities.
 - Soils are crucial for providing food, fiber, and other basic materials.
 - Soils are one of the most abused resources.

6.7 The Impact of Human Activities on Soil (cont.)

- Clearing the tropical rain forest: A Case Study
 - Tropical forests are cleared for logging and agricultural use.
 - Soils in tropical forests are poor in nutrients and unsuitable for agriculture.
 - Most of the nutrients in tropical rain forests are found in the trees.
 - Clearing tropical rain forests also promotes soil erosion.

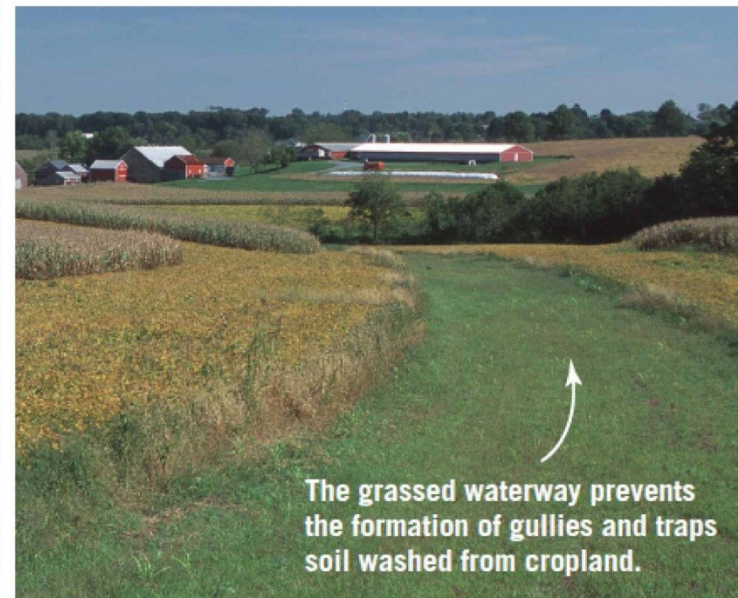
Tropical Deforestation



6.7 The Impact of Human Activities on Soil

- Soil erosion
 - Soil erosion is a natural process in the rock cycle.
 - Erosion rates are dependent on climate, slope, and type of vegetation.
 - Human activities such as deforestation and farming practices can enhance soil erosion.

Examples of Soil Conservation



6.8 Weathering & Ore Deposits

- Weathering creates deposits by concentrating metals into economically valuable concentrations (secondary enrichment).
- Bauxite
 - Principal ore of aluminum
 - Forms in rainy tropical climates from chemical weathering and the removal of undesirable elements by leaching

Bauxite



6.8 Weathering & Ore Deposits

- Other deposits
 - Weathering processes concentrate metals that are deposited through low-grade primary ore.
 - Examples: Copper and Silver
 - Typically occurs in deposits containing pyrite